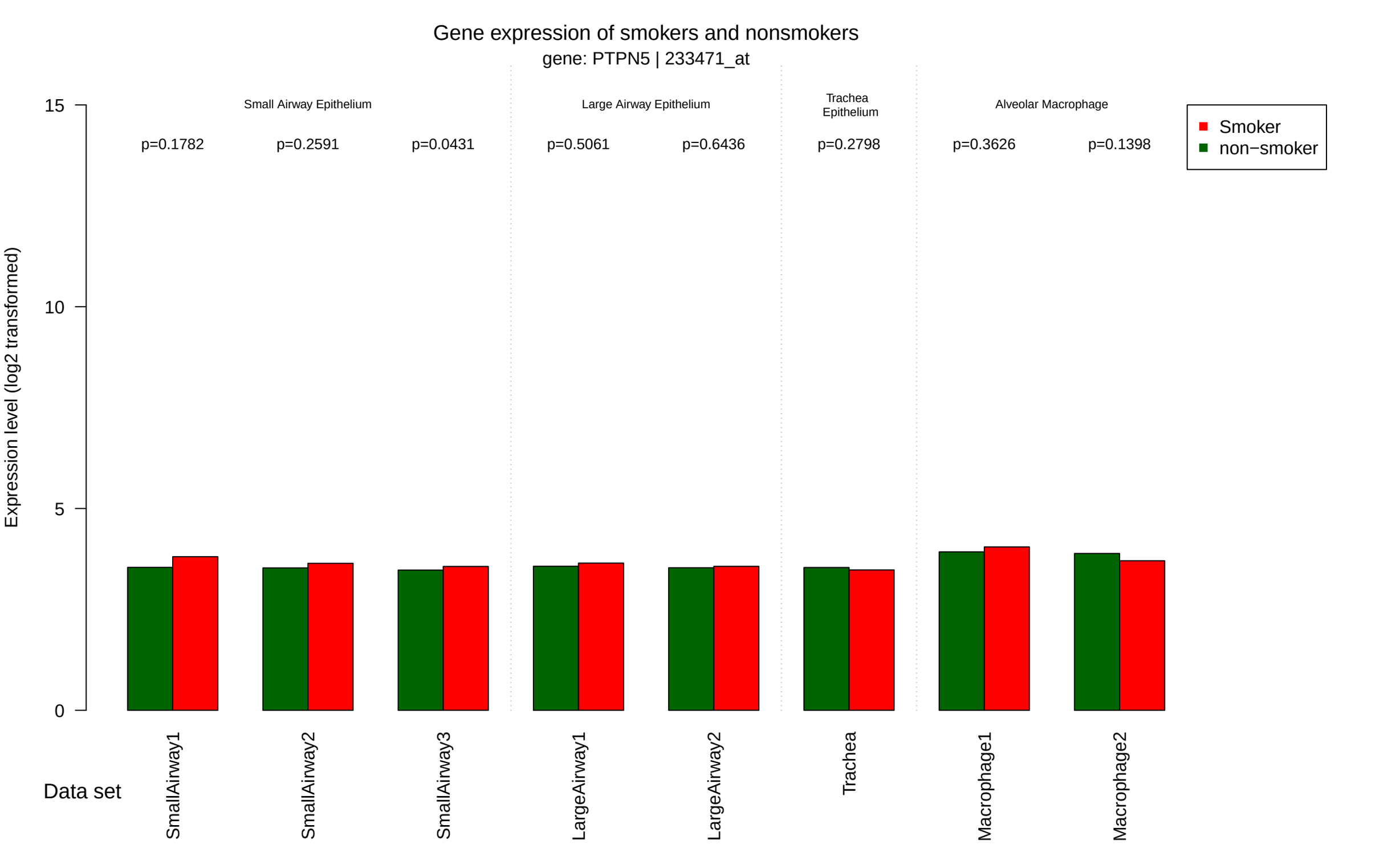
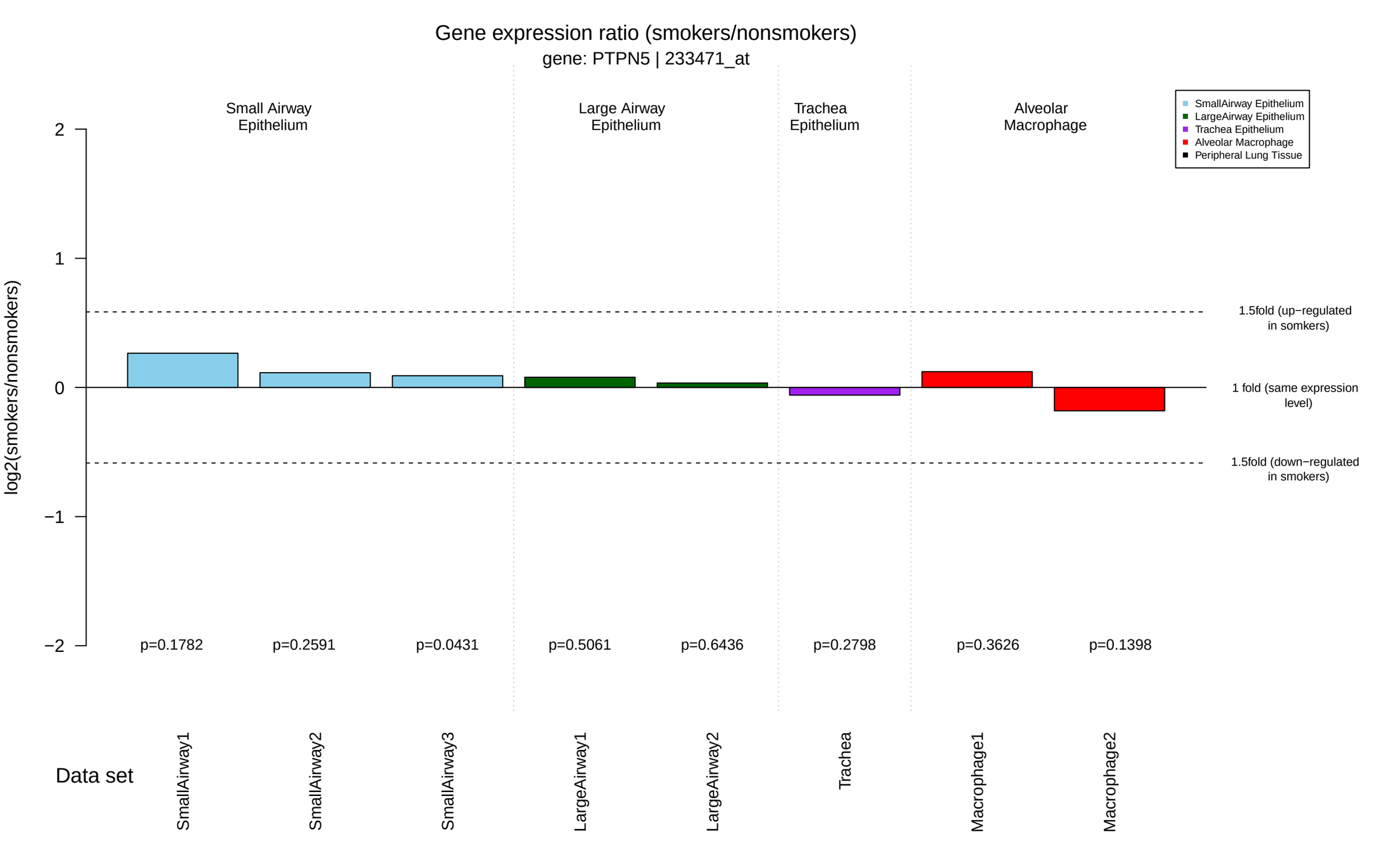
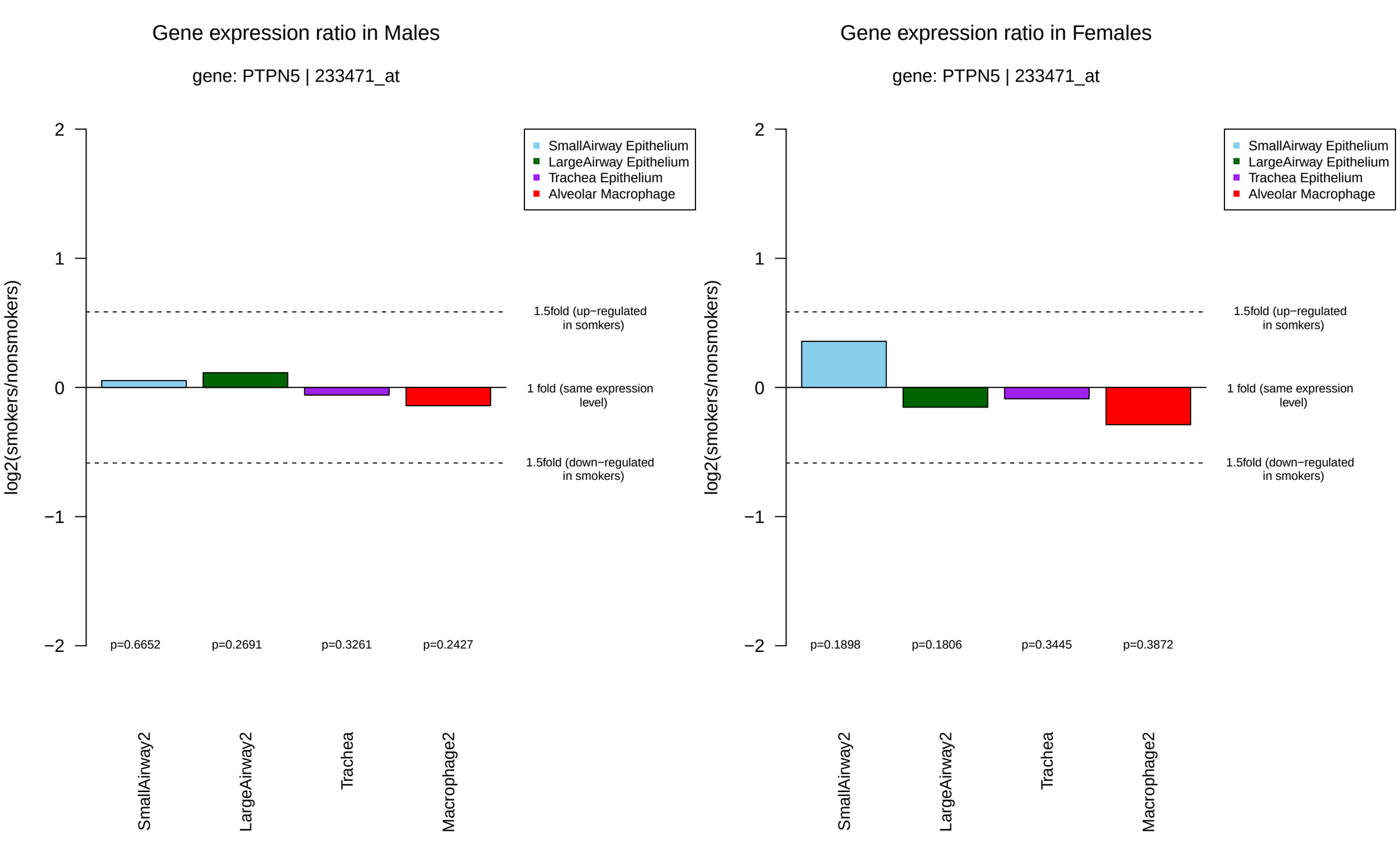
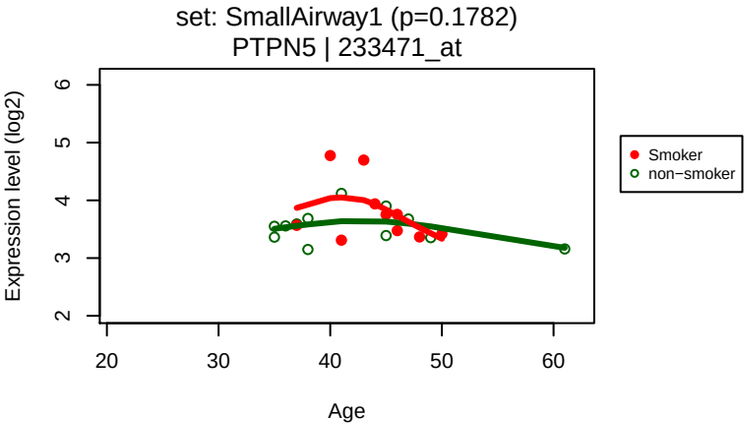
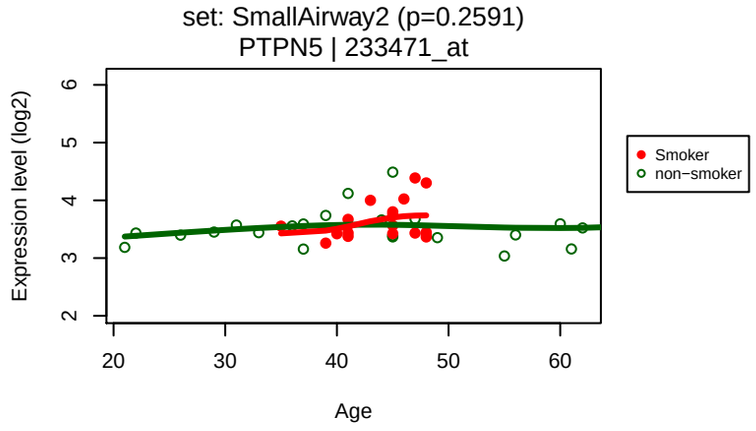
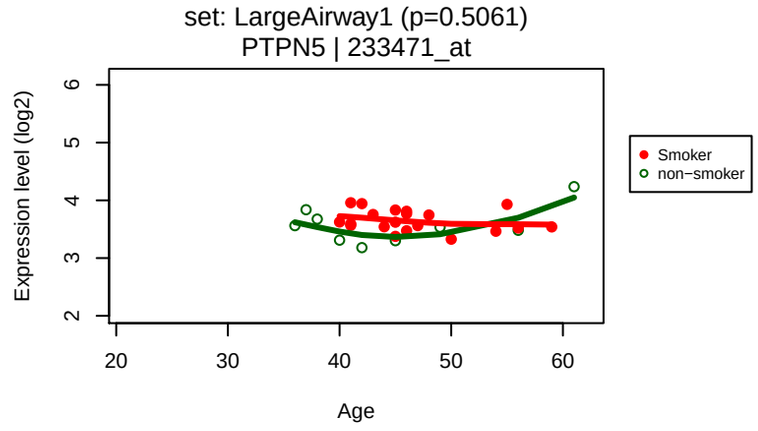

Gene expression of smokers and nonsmokers
gene: PTPN5 | 233471_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1782
p=0.2591
p=0.0431
p=0.5061
p=0.6436
p=0.2798
p=0.3626
p=0.1398
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

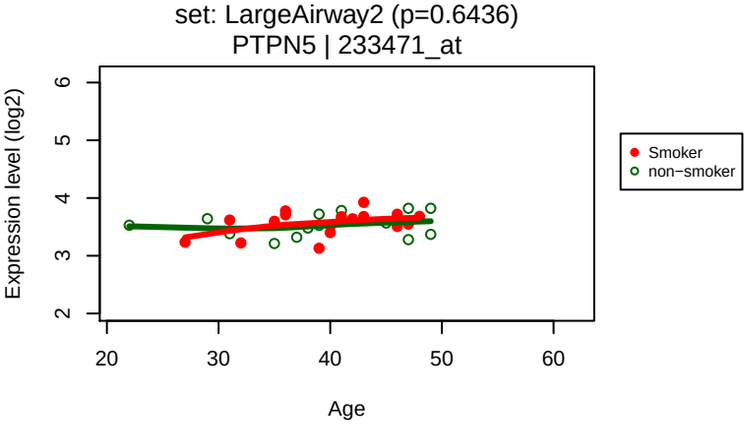

Gene expression ratio (smokers/nonsmokers)
gene: PTPN5 | 233471_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1782
p=0.2591
p=0.0431
p=0.5061
p=0.6436
p=0.2798
p=0.3626
p=0.1398
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

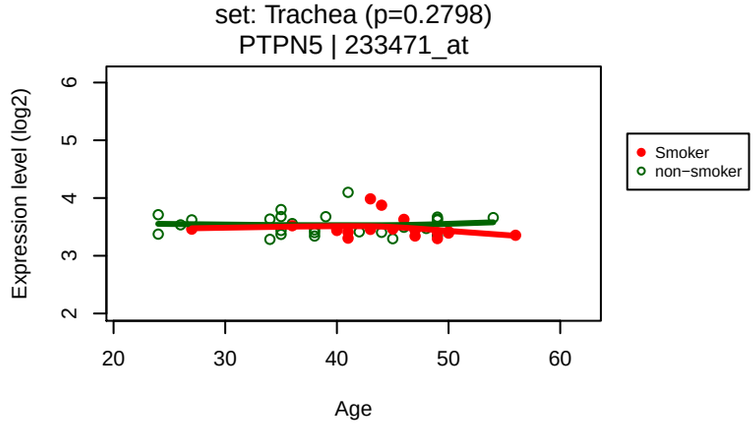

Gene expression ratio in Males
gene: PTPN5 | 233471_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6652
p=0.2691
p=0.3261
p=0.2427
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN5 | 233471_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1898
p=0.1806
p=0.3445
p=0.3872
SmallAirway2
LargeAirway2
Trachea
Macrophage2

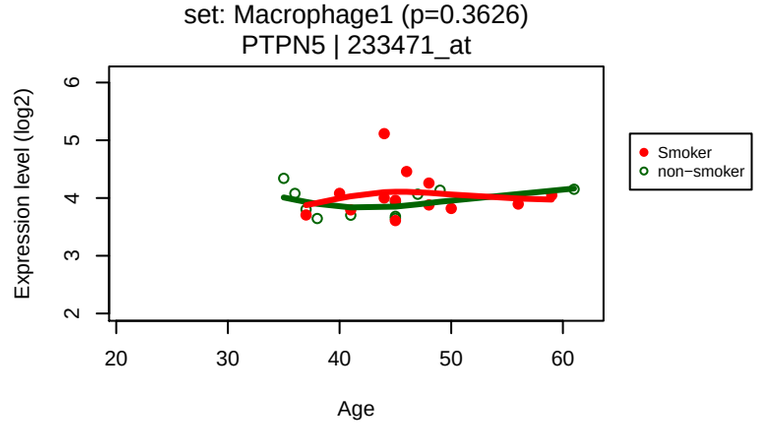

set: SmallAirway1 (p=0.1782)
PTPN5 | 233471_at
Expression level (log2)
Age
Smoker
non−smoker

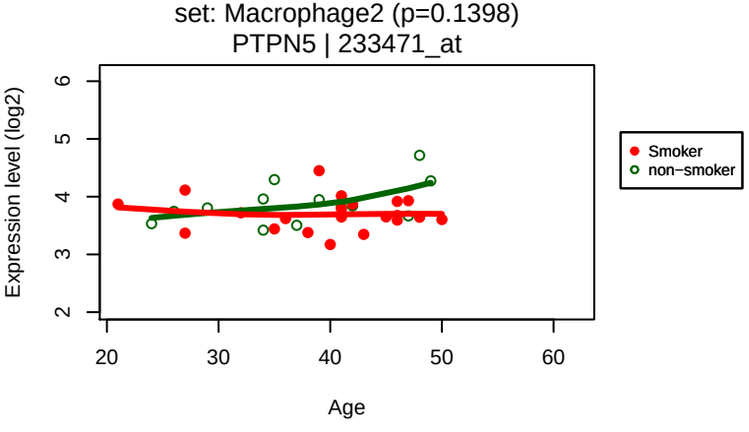

set: SmallAirway2 (p=0.2591)
PTPN5 | 233471_at
Expression level (log2)
Age
Smoker
non−smoker

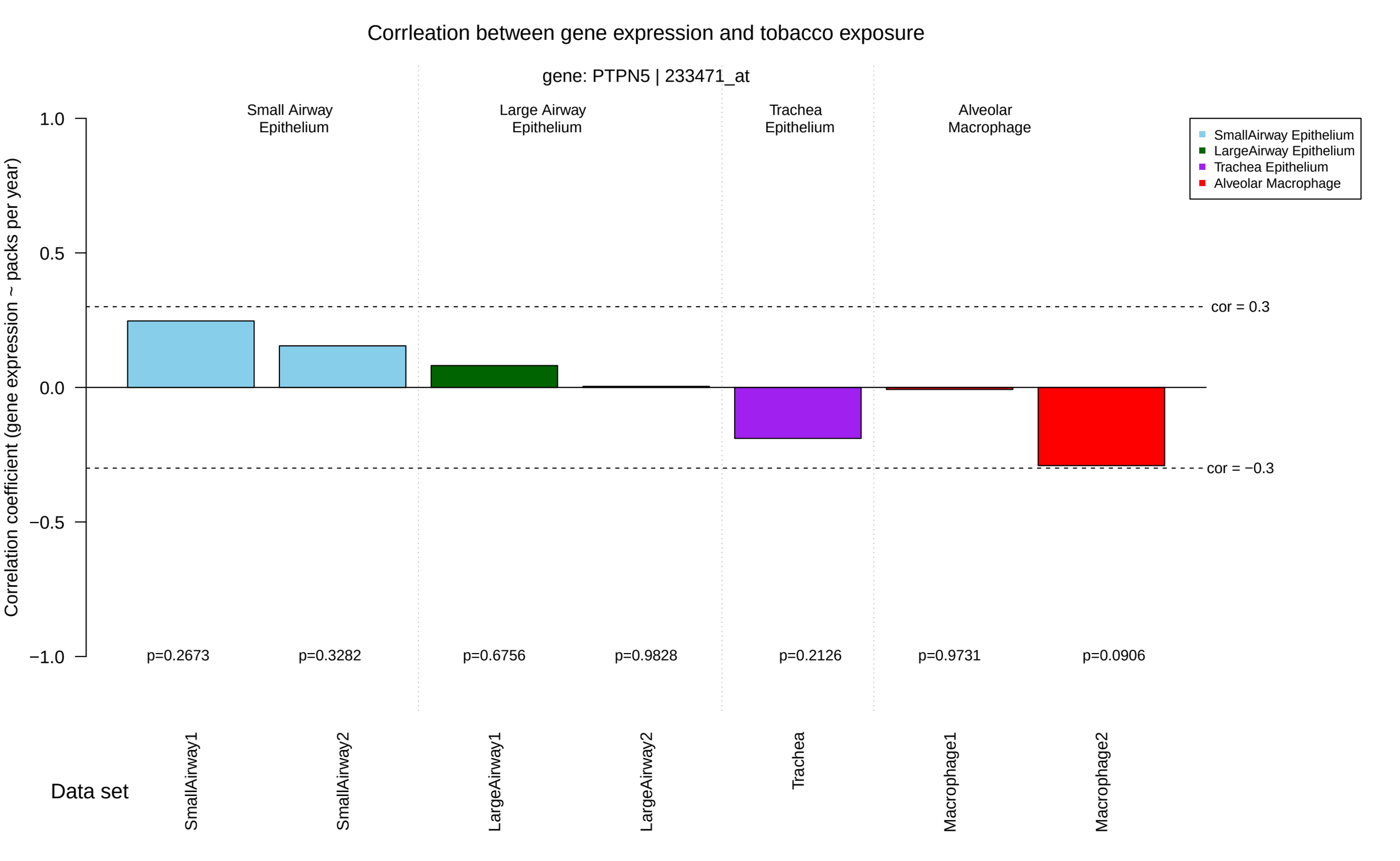

set: LargeAirway1 (p=0.5061)
PTPN5 | 233471_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6436)
PTPN5 | 233471_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2798)
PTPN5 | 233471_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3626)
PTPN5 | 233471_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1398)
PTPN5 | 233471_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN5 | 233471_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2673
p=0.3282
p=0.6756
p=0.9828
p=0.2126
p=0.9731
p=0.0906
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2